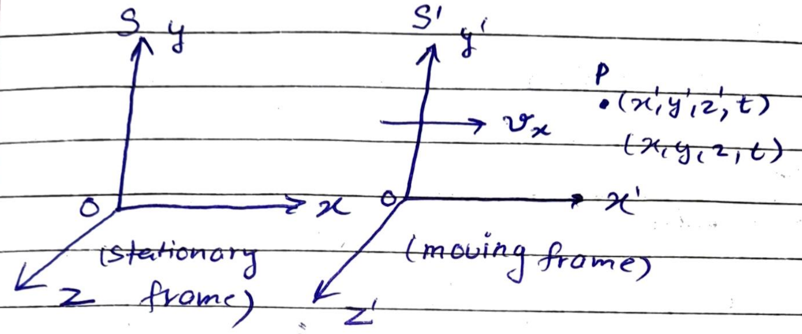
Tuesday
25/03/25

Physics

Unit - IV ; Theory of Relativity

→ Frame of reference – inertial / Non-inertial



- **Galilean Transformation Equation :-**

for x-direction	for y-direction
$x' = x - vt$	$x' = x$
$y' = y$	$y' = y - vt$
$z' = z$	$z' = z$
$t' = t$	$t' = t$

- **Postulates of Special Theory of Relativity :-**

1) All the fundamental laws of physics retain the same form in all the inertial frame of reference.

2) The velocity of light in free space is constant and is independent of the relative motion of the source & the observer.

- Lawrence Transformation Equation :-

↳ $x' = \gamma(x - vt)$ where, $\gamma = \dfrac{1}{\sqrt{1 - v^2/c^2}}$

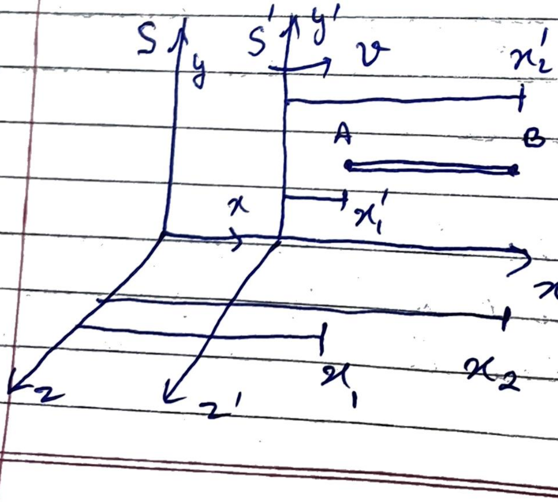
$$\underbrace{\begin{aligned} x' &= \gamma(x - vt) \\ y' &= y \\ z' &= z \\ t' &= \gamma\left(t - \frac{vx}{c^2}\right) \end{aligned}}_{\text{for } x\text{-direction}}$$

↳ $$\left.\begin{aligned} x' &= x \\ y' &= \gamma(y - vt) \\ z' &= z \\ t' &= \gamma\left(t - \frac{vy}{c^2}\right) \end{aligned}\right\} \text{for } y\text{-direction}$$

- Inverse - Lawrence Transformation Equation :-

↳ $$\left.\begin{aligned} x &= (x' + vt')\gamma \\ y &= y' \\ z &= z' \\ t &= \left(t' + \frac{vx'}{c^2}\right)\gamma \end{aligned}\right\} \text{for } x\text{-direction}$$

- Length Contraction :-



$L_0 = x_2' - x_1'$ — ①

(proper or actual length)

$L = x_2 - x_1$

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As per lawrence transformation equation,

$x_1' = (x_1 - vt)\gamma$ & $x_2' = (x_2 - vt)\cancel{\gamma}\gamma$

Put this in eq^n-(1),

$\Rightarrow L_0 = (x_2 - vt - x_1 + vt)\gamma = (x_2 - x_1)\gamma$

$\Rightarrow L_0 = \gamma L$

$\therefore L_0 = \dfrac{L}{\sqrt{1 - v^2/c^2}}$

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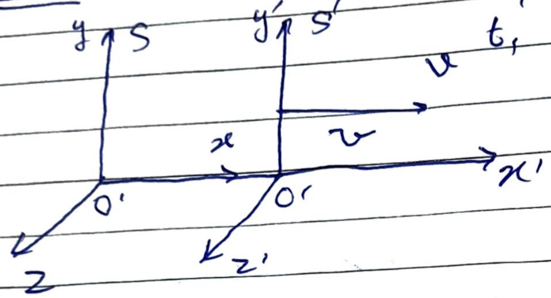
Physics

Theory of Relativity

- **Time dialation :-**



$t_1' \quad t_2'$ $\qquad t' = t_2' - t_1'$

$t_1 \quad t_2$ $\qquad t = t_2 - t_1$

So,

$$t_1 = \frac{t_1' + \frac{vx'}{c^2}}{\sqrt{1 - v^2/c^2}} \quad \text{— (1)}$$

$$t_2 = \frac{t_2' + \frac{vx'}{c^2}}{\sqrt{1 - v^2/c^2}} \quad \text{— (2)}$$

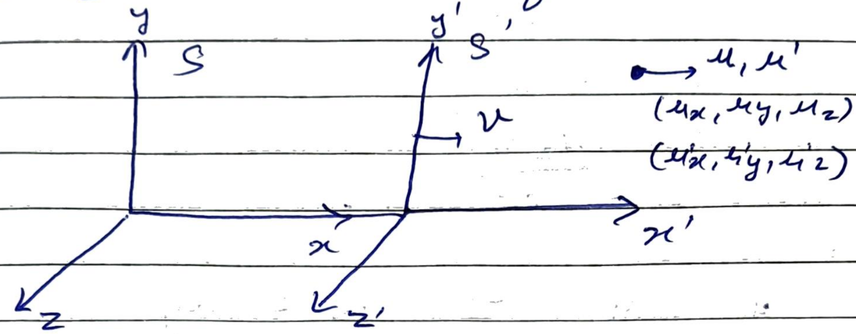
So, $t_2 - t_1 = \dfrac{t_2' - t_1'}{\sqrt{1 - v^2/c^2}}$

$\Rightarrow \quad t = \dfrac{t'}{\sqrt{1 - v^2/c^2}}$, Here $t > t'$.

- **Velocity Transformation Equation :-**

Consider two frames S and S' where

is moving towards x direction relative to frame S with velocity v. Suppose a particle moves with velocity u whose components are u_x, u_y, u_z relative to frame S and with velocity u' whose components are u_x', u_y' & u_z' relative to frame S'.



$$u_x = \frac{dx}{dt}, \quad u_y = \frac{dy}{dt} \quad \& \quad u_z = \frac{dz}{dt}$$

$$u_x' = \frac{dx'}{dt'}, \quad u_y = \frac{dy'}{dt'} \quad \& \quad u_z = \frac{dz'}{dt'}$$

Now,

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}} \Rightarrow dx' = \frac{dx - v\,dt}{\sqrt{1 - v^2/c^2}}$$

$$t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - v^2/c^2}} \Rightarrow dt' = \frac{dt - \frac{v\,dx}{c^2}}{\sqrt{1 - v^2/c^2}}$$

$$\Rightarrow u_x' = \frac{dx'}{dt'} = \frac{dx - v\,dt}{dt - v\,dx} = \frac{dx/dt - v}{1 - v\frac{dx}{dt}}$$

$$= \frac{u_x - v}{1 - \frac{v u_x}{c^2}}$$

$$\Rightarrow u_x' = \frac{u_x - v}{1 - \frac{u_x v}{c^2}}$$

Similarly,

$$u_y' = \frac{u_y - v}{1 - \frac{v u_y}{c^2}} \quad \&$$

$$u_z' = \frac{u_z - v}{1 - \frac{u_z v}{c^2}}.$$

Now,

$$y' = y \rightarrow dy' = dy$$

$$\& \quad t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - v^2/c^2}} \Rightarrow dt' = \frac{dt - \frac{v\,dx}{c^2}}{\sqrt{1 - v^2/c^2}}$$

$$\therefore \frac{dy'}{dt'} = u_y' = \frac{dy}{dt - \frac{v\,dx}{c^2}} \times \frac{1}{\gamma}$$

$$= \frac{dy/dt}{1 - \frac{v}{c^2}\frac{dx}{dt}} \times \frac{1}{\gamma}$$

$$u_y' = \frac{u_y}{\left(1 - \frac{v u_x}{c^2}\right)\gamma}$$

Similarly,

$$u_z' = \frac{u_z}{\left(1 - \frac{v u_x}{c^2}\right)\gamma}$$

And,

$$u_x = \frac{u_x' + v}{1 + \frac{u_x' v}{c^2}} \quad , \quad u_y = u_y' \left(1 - \frac{v u_x}{c^2}\right) \gamma$$

$$u_z = u_z' \left(1 - \frac{v u_x}{c^2}\right) \gamma$$

Thursday
3/04/25

Physics

Theory of Relativity

Q. At what speed should a clock be move so that it may appear to lose one minute in each hour?

Q. A rod has a length of 2m, find its length when it is carried in a rocket with a speed 2.7×10⁸ m/s.

Q. A spaceship moving away from the earth with velocity 0.6c fires a rocket whose velocity relative to the spaceship is 0.7c. What will be the velocity of the rocket as observed from earth in two cases
(i) Away from the earth
(ii) Towards the earth.

Sol^n-1) Here, if $t' = 59$ then $t = 60$

$$\therefore \quad t = \frac{t'}{\sqrt{1 - v^2/c^2}} \Rightarrow 60 = \frac{59}{\sqrt{1 - v^2/c^2}}$$

$$\Rightarrow \quad 1 - \frac{v^2}{c^2} = \left(\frac{59}{60}\right)^2$$

$$\Rightarrow \quad \frac{v^2}{c^2} = 1 - 0.96694 = 0.3305$$

$$\Rightarrow \quad v^2 = 0.3305\,c^2$$

$$\Rightarrow \quad v = 0.1818\,c$$

$$\Rightarrow \quad v = 5.454 \times 10^7\ m/s$$

Sol^n-2) $L_0 = \dfrac{L}{\sqrt{1-\frac{v^2}{c^2}}}$

$\Rightarrow 2 = \dfrac{L}{\sqrt{1-\left(\frac{2.7\times10^8}{3\times10^8}\right)^2}}$

$\Rightarrow L = 2\times0.43589$

$\Rightarrow L = 0.871\,m$

Sol^n-3) $u_x' = \dfrac{u_x - v}{1-\frac{u_x v}{c^2}}$

$\Rightarrow 0.7c = \dfrac{u_x - 0.6\cancel{c}}{1-\frac{u_x\times0.6\cancel{c}}{c\cancel{2}}}$

$\Rightarrow 1-\dfrac{u_x\times0.6}{c} = \dfrac{u_x - 0.6c}{0.7c}$

$\Rightarrow \dfrac{0.6u_x}{c} + \dfrac{u_x}{0.7c} = 1+\dfrac{6}{7}$

$\Rightarrow u_x\left(\dfrac{0.6}{c}+\dfrac{1}{0.7c}\right) = \dfrac{13}{7}$

$\Rightarrow \dfrac{u_x}{c}\left(0.6+\dfrac{1}{0.7}\right) = \dfrac{13}{7}$

$\Rightarrow \dfrac{u_x}{c}\times2.02857 = \dfrac{13}{7}$

$\Rightarrow \boxed{u_x = 0.9155c}$ [Away from Earth]

$u_x = 2.746\times10^8\,m/s$

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$$u_x' = \frac{u_x + v}{1 + \frac{u_x v}{c^2}} \Rightarrow 0.7c = \frac{u_x + 0.6c}{1 + \frac{u_x \times 0.6\cancel{c}}{c^{\cancel{2}}}}$$

$$\Rightarrow 1 + \frac{0.6u_x}{c} = \frac{u_x}{0.7c} + \frac{6}{7}$$

$$\Rightarrow \frac{u_x}{c}\left(0.6 + \frac{1}{0.7}\right) = \frac{1}{7}$$

$$\Rightarrow \frac{u_x}{c} \times \cancel{2.02857}\ 0.8285 = \frac{1}{7}$$

$$\Rightarrow u_x = 0.17c$$

$$\Rightarrow \boxed{u_x = 5.11 \times 10^7\ m/s}$$ [Towards the Earth]

Q. Half life of a particle at rest is 17.8 ns. What will be the half life when its speed is 0.8 c

Solⁿ→

$$t_{1/2} = \frac{t^\circ_{1/2}}{\sqrt{1 - v^2/c^2}}$$

$$\Rightarrow t_{1/2} = \frac{17.8\,ns}{\sqrt{1 - \left(\frac{0.8c}{c}\right)^2}} = \frac{17.8}{\sqrt{0.36}} = \frac{17.8}{0.6}$$

$$\Rightarrow \boxed{t_{1/2} = 29.67\ ns}$$

Q. The length of a rod is found to be half of its length when at rest. What is the speed of rod relative to the observer

Solⁿ→

$$l = \frac{l}{2\sqrt{1 - \frac{v^2}{c^2}}} \Rightarrow 1 - \frac{v^2}{c^2} = \frac{1}{4} \Rightarrow \frac{v^2}{c^2} = \frac{3}{4}$$

$$v = \frac{\sqrt{3}}{2}c$$

$$\Rightarrow v = 0.866c$$

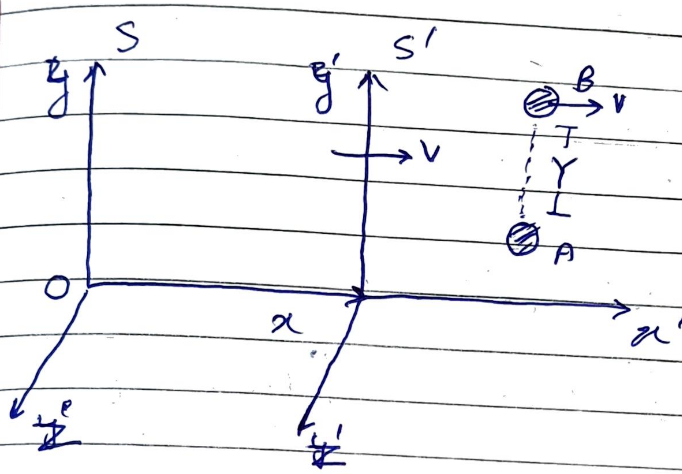
Monday
7/04/25

Physics

Theory of Relativity

- **Variation of mass with velocity :-**

$\hookrightarrow$ $$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

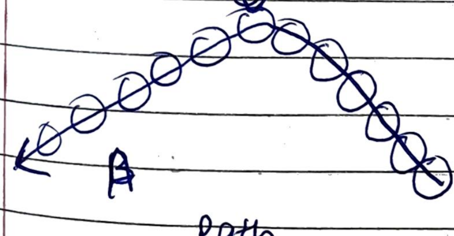


$m_A = m_B$

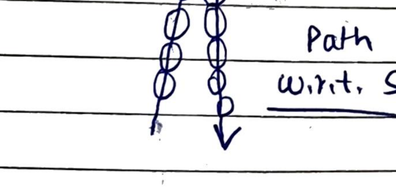
$V_{Ay} = V_{By}$

The collision will take place at $y = Y/2$

$\Rightarrow \therefore y' = Y/2$



Path w.r.t S'



Path w.r.t. S

For A, $T_0 = \frac{Y}{V_A}$ [From S]

& for B, $T_0 = \frac{Y}{V_B}$ [From S']

Also, $m_A V_{Ay} = m_B V_{By}$ —— ①

for B, $T' = \frac{Y}{V_B}$ [From S] & $T' = \frac{T_0}{\sqrt{1-v^2/c^2}}$

$$\Rightarrow \frac{Y}{V_{By}} = \frac{Y/V_{Ay}}{\sqrt{1-v^2/c^2}}$$

$$\Rightarrow V_{By} = V_{Ay}\sqrt{1-v^2/c^2}$$

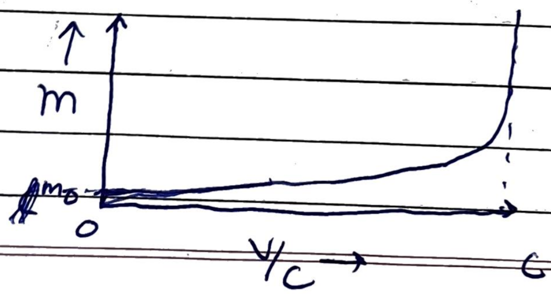
$$\Rightarrow \frac{V_{By}}{V_{Ay}} = \sqrt{1-v^2/c^2}$$ [From eqn-①]

$$\Rightarrow \frac{m_A}{m_B} = \sqrt{1-v^2/c^2}$$

$$\Rightarrow \boxed{m_B = \frac{m_A}{\sqrt{1-v^2/c^2}}}$$

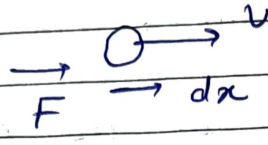
So if, $m_B = m$ (moving mass) &
$m_A = m_0$ (Rest mass)

then, $$m = \frac{m_0}{\sqrt{1-v^2/c^2}}$$



• Einstein's Mass - Energy Relation :-

↳ $E = MC^2$



$\Rightarrow F = \frac{dP}{dt}$

$\Rightarrow F dx = \frac{dP}{dt} dx \Rightarrow dW = \frac{dP}{dt} dx$

$\Rightarrow \therefore dk = \left(\frac{dx}{dt}\right) dP \Rightarrow dk = v dP$

$\Rightarrow dK = v d(mv) = v(mdv + vdm)$

$\Rightarrow dk = mv dv + v^2 dm$ ——— ①

Now, $m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$

$\Rightarrow m^2\left(1 - \frac{v^2}{c^2}\right) = m_0^2$

$\Rightarrow m^2c^2 - m^2v^2 = m_0^2c^2$

$\Rightarrow 2m\,dm\,c^2 - (2m\,dm\,v^2 + 2v\,dv\,m^2) = 0$

$\Rightarrow dm\,c^2 = mv\,dv + v^2 dm$ ——— ②

So, $dk = dm\,c^2$ [From eqⁿ ①]

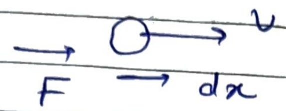
$\Rightarrow \int_0^K dk = \int_{m_0}^{m} dm\,c^2 \Rightarrow K = (m - m_0)c^2$

$\Rightarrow \underbrace{K} + \underbrace{m_0c^2} = mc^2$

Kinetic energy Potential energy (rest mass energy) eisenber

Einstein's Mass - Energy Relation :-

$\hookrightarrow E = Mc^2$



$\Rightarrow F = \dfrac{dP}{dt}$

$\Rightarrow F dx = \dfrac{dP}{dt} dx \Rightarrow dW = \dfrac{dP}{dt} dx$

$\Rightarrow \therefore dk = \left(\dfrac{dx}{dt}\right) dP \Rightarrow dk = v\,dP$

$\Rightarrow dk = v\,d(mv) = v(m\,dv + v\,dm)$

$\Rightarrow dk = mv\,dv + v^2 dm$ ——— (1)

Now, $m = \dfrac{m_0}{\sqrt{1 - v^2/c^2}}$

$\Rightarrow m^2\left(1 - \dfrac{v^2}{c^2}\right) = m_0^2$

$\Rightarrow m^2c^2 - m^2v^2 = m_0^2c^2$

$\Rightarrow 2m\,dm\,c^2 - (2m\,dm\,v^2 + 2v\,dv\,m^2) = 0$

$\Rightarrow dm\,c^2 = mv\,dv + v^2 dm$ ——— (2)

So, $dk = dm\,c^2$ [From eq^n (1)]

$\Rightarrow \displaystyle\int_0^k dk = \int_{m_0}^{m} dm\,c^2 \Rightarrow k = (m - m_0)c^2$

$\Rightarrow \underbrace{k}_{\text{Kinetic energy}} + \underbrace{m_0c^2}_{\text{Potential energy (rest mass energy)}} = mc^2 \Rightarrow$ Total Energy$(E) = mc^2$